

FixGrid: Decentralized Trust & Warranty Infrastructure

Rewiring the \$15.2B Local Repair Economy Through Milestone Smart Escrow & Platform Guarantees

Live Platform:
www.vytron.me

Architecture: Next.js 16 (App Router) • Supabase
PostgreSQL • Leaflet Engine

Mission: Right-to-Repair & Zero-Waste
Circular Economy

01. EXECUTIVE SUMMARY

Bridging Everyday Consumers with Neighborhood Repair Artisans

Every year, millions of smartphones, home appliances, laptops, and wearables are discarded prematurely. Consumers spend hundreds of billions of rupees replacing items that could be repaired for less than 15% of replacement cost. This throwaway cycle is not fueled by laziness—it is driven by an acute **Trust and Information Asymmetry**. Consumers fear arbitrary price gouging, counterfeit parts, component swapping, and zero post-repair warranty. Simultaneously, skilled neighborhood technicians—the true frontline artisans of the circular economy—remain digitally invisible and economically marginalized.

FixGrid is a full-stack trust infrastructure that solves this market failure. By combining a **Milestone-Based Smart Escrow Engine**, a **Platform-Backed 5-Day Guarantee (FixGrid Shield)**, and **Verified Digital Storefronts with Transparent Rate Cards**, FixGrid standardizes fragmented local repairs into an accountable, trustworthy service experience as dependable as buying new. Crucially, FixGrid is not an unverified concept: it is **fully functional and live in production at www.vytron.me**.

02. THE PROBLEM STATEMENT

The Broken Economics of Repair & The Climate Paradox

84%

CONSUMER FEAR

The Lemon Market Trap

Consumers avoid local repair due to fear of arbitrary price gouging, fake counterfeit components, component theft, and zero warranty recourse. They default to buying expensive new replacements.

92%

ARTISAN INVISIBILITY

The Invisible Craftsman

Master repair technicians operate < 500m away, but lack any web presence, customer booking tools, or verified reputation ledger. They are starved of footfall and trapped by predatory middlemen.

1.71M+

TONNES OF E-WASTE / YEAR

The Climate Catastrophe

India is the #3 global e-waste generator. 80% of device lifetime carbon is emitted during manufacturing. Discarding repairable items causes massive toxic waste and accelerates climate breakdown.

03. PRODUCT ARCHITECTURE

FixGrid: The 4 Pillars of Pure Software Trust

PILLAR 1

Geospatial Discovery & Digital Identity

Custom Leaflet-powered geospatial engine mapping verified neighborhood repair workshops. Filter by exact device model, issue category, proximity, and customer reviews with transparent rate cards.

PILLAR 2

Milestone-Based Smart Escrow

Consumer funds are locked in the FixGrid Escrow Vault at booking. Funds are never disbursed prematurely; they are released only after the customer inspects the repaired device and inputs a completion OTP.

PILLAR 3

FixGrid Shield (Platform Warranty)

Universal 5-day platform-backed warranty on workmanship + recorded merchant extended warranties (30/90/180 days). Integrated dispute arbitration protocol with 100% money-back guarantee.

PILLAR 4

Shop Pro SaaS & 5% Cashback Rebate

5% completed-bill cashback rebate incentivizes offline shops to log all walk-in jobs on FixGrid. Optional ₹999/mo SaaS tier adds cloud inventory, job ticketing, and automated WhatsApp repair progress alerts.

04. ENGINEERING SPECIFICATIONS

Enterprise Web Architecture Built for Speed, Scale & Security

Component	Implementation & Architectural Advantages
Frontend & Geo-Engine	Next.js 16 (App Router) with React Server Components (RSC), TypeScript, and Tailwind CSS. Integrated Leaflet / OpenStreetMap engine delivers sub-second proximity search without heavy third-party API costs.
Backend & Data Security	Supabase (PostgreSQL) utilizing strict relational schemas and Row Level Security (RLS) for multi-tenant data isolation. Automated Database Triggers orchestrate escrow state machines and cryptographic completion OTP verification.
Programmatic SEO CMS	Dedicated /seo-admin headless CMS that programmatically synthesizes thousands of localized, long-tail keyword URLs (e.g. 'MacBook battery replacement in [Locality]'), delivering zero-CAC organic customer acquisition.

05. BUSINESS ECONOMICS

Monetizing a \$15.2B High-Frequency Unorganized Market

\$15.2B TOTAL ADDRESSABLE MARKET Indian unorganized repair sector across electronics, home appliances, and daily wearables (surging past \$25B with Right-to-Repair mandates).	\$4.8B SERVICEABLE MARKET Tier-1 and Tier-2 urban consumer electronics, smartphone, and appliance repair volume where digital payment adoption is > 80%.	\$120M 3-YEAR TARGET SOM Addressable urban hyperlocal repair GMV captured across initial metro clusters (Delhi NCR, Bengaluru, Mumbai).
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Revenue Stream	Mechanism & Economics
Platform Take-Rate (5% – 8%)	Transaction commission fee on each escrow-backed completed booking. Covers payment processing, escrow vault infrastructure, and the FixGrid Shield warranty pool.
Shop Pro SaaS (₹999 / Month)	Optional recurring subscription for busy workshops: multi-technician job ticketing, inventory alerts, automated customer WhatsApp repair status updates, and priority search placement.
Certified Spare Parts Logistics	B2B supply-chain marketplace (parts.vytron.me) connecting member shops directly with verified OEM-grade component suppliers at a 10%–18% distribution margin.

06. MEASURABLE IMPACT (ESG)

Triple Bottom Line: People, Planet & Livelihoods

Dimension	Core Metric	Societal & Planetary Significance
Economic (Consumer)	60% – 80% Savings	Substantial cost reduction vs. buying new replacements; unlocks ~₹14,000 in annual household savings and extends product life by 2.5 to 4 years.
Social (Artisans)	+35% – 45% Income	Formalizes unorganized street technicians into verified micro-entrepreneurs. Immutable rating ledger unlocks access to formal microcredit and banking.
Ecological (Planet)	72 kg CO2e Saved	Greenhouse emissions prevented per repaired smartphone by eliminating manufacturing emissions. Diverts toxic e-waste directly from landfills.

07. HACKATHON WINNING EDGE

Tested, Deployed & Operational in Production

FULLY DEPLOYED IN PRODUCTION

Live Platform: www.vytron.me

Unlike conceptual hackathon prototypes, FixGrid is live today. Explore the Leaflet map, verified artisan storefronts, transparent rate cards, and responsive mobile/desktop UI right now.

Visit www.vytron.me